



Paper Id: 252744

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Subject Code: BMBA1205

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MBAINT
(SEM II) THEORY EXAMINATION 2024-25
BASICS OF STATISTICS

TIME: 3 HRS

M.MARKS: 70

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.

02 x 7 = 14

Q no.	Question
a.	What is meant by Frequency Distribution?
b.	What do you mean by Kurtosis?
c.	What is Quartile Deviation?
d.	Define the relationship between Correlation and Regression Coefficients.
e.	What is meant by Secular Trend?
f.	Define the Additive and Multiplicative Models of Time Series.
g.	Define Family Budget Method.

SECTION B

2. Attempt any three of the following:

07 x 3 = 21

Q no.	Question																																									
a.	What is Statistics? Discuss the classification of data through suitable examples.																																									
b.	What do you mean by Measures of Central Tendency? Discuss the various Parts of Measures of Central Tendency;																																									
c.	From the following table, calculate the Karl Pearson's Coefficient of Correlation between age of husbands and wives:- <table border="1" style="margin: 10px auto;"><thead><tr><th rowspan="2">Age of Wives (Years)</th><th colspan="5">Age of Husbands (Years)</th></tr><tr><th>20-30</th><th>30-40</th><th>40-50</th><th>50-60</th><th>60-70</th></tr></thead><tbody><tr><td>55-65</td><td>-</td><td>-</td><td>-</td><td>4</td><td>2</td></tr><tr><td>45-55</td><td>-</td><td>-</td><td>4</td><td>16</td><td>5</td></tr><tr><td>35-45</td><td>-</td><td>1</td><td>12</td><td>2</td><td>-</td></tr><tr><td>25-35</td><td>-</td><td>10</td><td>25</td><td>2</td><td>-</td></tr><tr><td>15-25</td><td>5</td><td>9</td><td>3</td><td>-</td><td>-</td></tr></tbody></table>	Age of Wives (Years)	Age of Husbands (Years)					20-30	30-40	40-50	50-60	60-70	55-65	-	-	-	4	2	45-55	-	-	4	16	5	35-45	-	1	12	2	-	25-35	-	10	25	2	-	15-25	5	9	3	-	-
Age of Wives (Years)	Age of Husbands (Years)																																									
	20-30	30-40	40-50	50-60	60-70																																					
55-65	-	-	-	4	2																																					
45-55	-	-	4	16	5																																					
35-45	-	1	12	2	-																																					
25-35	-	10	25	2	-																																					
15-25	5	9	3	-	-																																					
d.	What is meant by Time Series? Discuss the Moving Average Method for measurement of trend.																																									
e.	What do you mean by Time Reversal and Factor Reversal Tests? Discuss these tests in case of Ideal Index Number.																																									

SECTION C

3. Attempt any one part of the following:

07 x 1 = 07

Q no.	Question							
a.	Draw the Histogram for the following frequency distribution:-							
	Wages per day (in Rupees)	2-4	4-8	8-14	14-16	16-20	20-24	24-26
	No. of Workers	3	56	144	41	52	32	2

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b.	Prepare a frequency distribution of the following marks (maximum marks being 100) obtained by 60 students in a test taking 10 class intervals as: 0-10, 10-20,																		
	15	63	41	55	72	42	32	23	30	48	67	34							
	65	11	17	60	53	93	57	03	29	31	82	50							
	58	35	83	33	92	43	66	41	78	72	33	32							
	60	82	63	40	88	80	24	60	09	58	44	74							
	99	10	54	37	62	00	64	30	31	63	57	60							

4. Attempt any **one** part of the following: 07 x 1 = 07

a.	What is meant by Skewness? Discuss the Measurement of Skewness.																		
b.	For a distribution based on 200 observations partly reproduced below, mean is 1.46, find the missing frequencies:-																		
	No. of Accidents:	0	1	2	3	4	5												
	Frequency	: 46	?	?	25	10	5												

5. Attempt any **one** part of the following: 07 x 1 = 07

a.	Differentiate between Correlation & Regression Analysis.																		
b.	Find the two regression lines from the following data:-																		
	X:	11	14	14	17	17	21	25											
	Y:	15	27	27	30	34	38	46											
	Also estimate the value of Y for X = 40																		

6. Attempt any **one** part of the following: 07 x 1 = 07

a.	What is Semi-average Method of Time Series Analysis? Discuss the Short term / time movements / fluctuations / variations of Time Series.																		
b.	From the following Time Series obtain trend value by three-yearly moving averages:																		
	Year	Sales (in '000Rs.)				Year	Sales (in '000Rs.)												
	2015	8				2020	12												
	2016	12				2021	16												
	2017	10				2022	17												
	2018	13				2023	14												
	2019	15				2024	17												

7. Attempt any **one** part of the following: 07 x 1 = 07

a.	What is Index Number? Discuss the Problems in the Construction of Price Index Numbers.																		
b.	Compute Laspeyre's, Paasche's and Marshall-Edgeworth's price index numbers from the following data:																		
	Item	Base Year				Current Year													
		Price		Quantity		Price		Quantity											
	A	2		20		5		15											
	B	4		4		8		5											
	C	1		10		2		12											
	D	5		5		10		6											

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